

ALEXANDRE SANMARTÍN GOYANES

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EDUCATION

Master of Science in Mechanical Engineering **Sep 2025 – Current**
ETH Zürich – Top 6 World Mechanical Engineering (QS); Top 7 World Overall (QS) *Zürich, Switzerland*
• **Focus:** Aerospace Technologies, Product Design, and Global Production.

Bachelor of Science in Mechanical Engineering (International Degree) **Oct 2022 – Aug 2025**
Karlsruhe Institute of Technology (KIT) *Karlsruhe, Germany*
• **GPA:** 2.0/3.7 (German Scale/ American) – Top 45 World Mech. Eng. (QS); Top 3 Germany (QS)
• **Awards:** **Schroff Foundation Scholarship** (Awarded to the single top performing student for excellence).

PROFESSIONAL EXPERIENCE

Karlsruhe Institute of Technology (KIT) **Karlsruhe, Germany**
Tutor Assistant – Institute of Mechanics, & Thermodynamics *Oct 2024 – Aug 2025*
• Selected by faculty board as a top 1% student to teach **Advanced Mathematics, Thermodynamics, and Mechanics**.
• Instructed 30+ undergraduate students weekly, simplifying complex engineering concepts which led to a measurable improvement in group pass rates. Enhanced personal communication and leadership skills

ADDITIONAL EXPERIENCE & LEADERSHIP

KA-Raceing e.V. (Formula Student) **Karlsruhe, Germany**
Junior Engineer – Hydrogen Combustion Division *Oct 2024 – Apr 2025*
• **Platform Revival:** Revitalized a legacy 2019 chassis to prototype a new Hydrogen Combustion vehicle, managing the full restoration of missing components and system integration.
• **System Engineering:** Developed the hydrogen drivetrain from scratch, including the design of high-pressure pipelines, tank integration, valves, and safety systems.
• **Testing & Validation:** Executed engine tuning for hydrogen fuel and led safety certification processes, successfully passing the structural audit and Center of Gravity (tilt) tests. First Position in Overall, Business Plan, Engineering Design and Cost Report

Industrial Fendt Vario Tractor Design **Academic Project – 2024**
• Designed critical mechanical subsystems for a heavy-duty Fendt Vario tractor, including a **portal axle, planetary in-wheel gearbox, and differential**.
• Engineered auxiliary machine interfaces and a speed-selecting gearbox, optimizing the assembly for agricultural operational loads.
• Modeled complex multi-system interactions in CAD to ensure manufacturing feasibility and durability.

Bachelor Thesis: Global Production Optimization **Karlsruhe, Germany – Feb 2025 – Aug 2025**
• Developed a multivariable regression model in **Python** ($R^2 > 0.95$) to identify correlations between machine deviations and product defects.
• Processed and analyzed large manufacturing datasets to pinpoint error sources, enabling targeted improvements in production quality.
• Synthesized analytical findings into a comprehensive report on advanced manufacturing optimization.

LANGUAGES, IT SKILLS & INTERESTS

- **Languages:** Spanish & Galician (Native); **English** (Bilingual/C2); **German** (B2); **French** (DELFI B2).
- **IT & Engineering Skills:** Python, MATLAB, SQL, JavaScript (OOP).
- **Software:** CREO, Siemens NX, AutoCAD, Microsoft Excel (Advanced), AI Tools.
- **Interests:** Automotive Engineering (Hydrogen/EV), Sailing, Golf, Trumpet (Conservatory trained), Competitive Debate.